Forest Loss Pressure on Indigenous Lands with Uncontacted Peoples in the Brazilian Amazon: An Open-Data Geospatial Assessment (2000–2023)

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Preprint — October 2026

Abstract. Brazil hosts the world's largest concentration of uncontacted (isolated) Indigenous peoples, yet no published quantitative assessment measures how much forest loss occurs inside and around the Indigenous Lands that host them, as distinct from Indigenous lands in general. Using only open data — official FUNAI territory boundaries (657 lands), mapped isolated-peoples occupation areas and records from the OPI observatory (115 records; 29 state-confirmed), and the Hansen/UMD Global Forest Change dataset (2000–2023, canopy cover >30%) — we compare forest loss inside 81 Indigenous Lands with isolated-peoples presence (TIPI) against (i) 155 matched non-isolated Indigenous Lands and (ii) surrounding 0–10 km and 10–50 km buffer rings. Median loss of year-2000 tree cover inside TIPI was 1.0%, versus 3.7% in the 0–10 km ring and 5.3% in the 10–50 km ring (Wilcoxon signed-rank $p < 10^{-10}$, large effects); TIPI also lost significantly less forest than matched controls (2.5% median; Mann–Whitney $p < 10^{-4}$). Pressure is strongly spatially concentrated (Gini 0.63): the ten highest-pressure territories account for nearly half (48%) of aggregate surrounding loss, led by Uru-Eu-Wau-Wau (222,000 ha of ring loss; 35% of its own tree cover lost) and Arariboia (112,000 ha; 48% ring loss share). Ring pressure was associated with distance to cities ($\rho = -0.46$, $p < 10^{-4}$) but, notably, not with distance to highways ($p = 0.16$), suggesting that settlement proximity and enclaves — not only trunk-road corridors — drive exposure. These results quantify, for the first time at territory level, the buffering function of uncontacted peoples' lands and identify where Brazil's protection apparatus — audited in 2024 as under-staffed and leaving 12 records without any territorial protection — is most urgently needed. All code, data provenance, and statistics are fully reproducible from the open repository. The study involves no field contact and publishes no settlement locations, in line with the no-contact policy for isolated peoples.

Keywords: uncontacted peoples; isolated peoples; *povos indígenas isolados*; Indigenous Lands; deforestation; Brazilian Amazon; forest cover change; protected areas; geospatial analysis

1 Introduction

Uncontacted peoples — in Brazil's official terminology, *povos indígenas isolados* — are indigenous groups who maintain no sustained relations with the surrounding national society [1, 2]. Brazil holds the world's largest number of such peoples: FUNAI, the National Foundation of Indigenous Peoples, maintained 114 official records of isolated peoples in 2024, of which the state had confirmed 29 and left twelve without any territorial protection instrument whatsoever [3]; a 2025 global survey estimates 124 groups in Brazil alone, roughly two-thirds of the world's 196 [2]. For these peoples, forest loss is not only an environmental problem: they depend entirely on the forest for food, medicine, and material culture, they have no acquired immunity to epidemic disease, and their isolation is itself a survival strategy — formally protected since 1987 by FUNAI's no-contact policy [1]. The epidemiological record underlines what is at stake: across 117 recorded contact-related epidemics in Greater Amazonia between 1875 and 2008, median mortality ranged between 18% and 20% per epidemic [4].

A large and credible literature establishes that Indigenous Lands in the Brazilian Amazon, in general, inhibit deforestation and fire [5], shelter over half of the region's remaining forest carbon [6], and — when backed by full collective property rights — reduce clearing by around two-thirds relative to comparable boundaries [7–9]. Roads and settlement expansion are consistently identified as the leading proximate drivers of pressure [10–12]. Yet this literature treats

Indigenous Lands as a single category. The lands that host *uncontacted* peoples are different in three ways that matter for both science and policy: their inhabitants cannot be consulted, relocated, or compensated; the state's protection instrument for them (restrictions-of-use ordinances and exclusion belts, monitored by the Coordination for Isolated and Recently Contacted Peoples, CGiirc) is administratively fragile — FUNAI's own 2024 internal audit found outdated registries, an authorised staffing deficit of 74 out of 156 field posts, and twelve records with no protection at all [3]; and any forest loss that reaches them carries existential, non-mitigable risk.

Despite this, no published study quantifies, at the level of individual territories and with matched controls, how much forest loss occurs inside and around the Indigenous Lands that host uncontacted peoples. Existing quantitative work either aggregates all Indigenous Lands [5, 13], works at basin scale across Indigenous Territories and Protected Areas jointly [14], or analyses specific road corridors [12]. Scholarship specifically on isolated peoples is overwhelmingly descriptive, legal, or advocacy-oriented; a 2025 global report documents threats qualitatively (logging, mining, agribusiness, missions) [2], and Brazil's official record remains fragmented across FUNAI spreadsheets, civil-society registries, and protection ordinances [3, 15]. The gap is consequential: if uncontacted peoples' lands are, in effect, the strongest forest-buffers in the Amazon — or, alternatively, quietly losing their buffering capacity along specific frontier segments — nobody currently knows.

This paper closes that gap with a fully open, reproducible geospatial assessment. We assemble three public data layers — official FUNAI territory boundaries, the mapped occupation areas and records of isolated peoples from the OPI observatory [15], and the Hansen/UMD Global Forest Change dataset [16, 17] — and measure forest-cover loss (2000–2023) inside 81 Indigenous Lands with isolated-peoples presence, against two complementary references: 155 area-matched non-isolated Indigenous Lands, and concentric buffer rings (0–10 km and 10–50 km) around each territory. We further relate surrounding pressure to accessibility (distance to mapped highways and to cities) and quantify its concentration across territories. The study uses no field data, involves no contact of any kind, and publishes no settlement locations (Section 8).

The contributions are fourfold. **(C1)** A transparent, scripted data-integration pipeline that reconciles the official territory registry with the isolated-peoples record system and versioned forest-change rasters, released as open code. **(C2)** The first territory-level quantitative assessment of forest-loss pressure on Brazil's uncontacted-peoples lands, relative to matched controls and buffer rings, with effect sizes and uncertainty. **(C3)** A ranked, evidence-based exposure list identifying the specific territories where surrounding pressure is highest — directly usable for protection prioritisation, and consistent with FUNAI's audited protection deficits. **(C4)** A documented, replicable ethics protocol for studying isolated peoples without endangering them.

2 Related Work

2.1 Indigenous Lands and deforestation in the Amazon

The protective effect of Indigenous Lands on Amazonian forests is among the best-documented findings in land-systems science. Nepstad et al. [5] showed that parks and Indigenous lands inhibited deforestation and fire across the Brazilian Amazon, with Indigenous lands — covering one-fifth of the region — acting as the region's most important barrier. Soares-Filho et al. [6] projected that the protected-area and Indigenous-land network, holding 54% of remaining forests, would avoid large carbon emissions under business-as-usual scenarios. Methodological work, however, showed that naive comparisons overstate protection: protected lands are preferentially located in remote, low-pressure areas, and matched-control designs shrink the estimated effect [18, 19]. Nolte et al. [13] found that strictly protected areas and Indigenous lands avoided the most deforestation under matching, while Blackman et al. [8] and Baragwanath and Bayi [7], Baragwanath et al. [9] established quasi-experimental evidence that formal tenure (titling in Peru; homologation in Brazil) substantially reduces clearing and can trigger regrowth. Garnett

et al. [20] placed Indigenous lands globally at ∼38 million km^2 of managed or tenured land, and Duarte Ritter et al. [14] showed that across the Amazon basin, land within Indigenous Territories and Protected Areas is far less impacted by human activity (14–16%) than unprotected land (38%). None of these studies separates the lands hosting uncontacted peoples.

2.2 Drivers of pressure: roads, small clearings, and frontier dynamics

Roads are the canonical driver of Amazonian deforestation: Barber et al. [10] quantified the mitigating effect of protected areas along roads, and the BR-319 corridor literature documents how paving plans alone reorganise pressure on Indigenous lands, including those with isolated peoples [11, 12]. Kalamandeen et al. [21] showed that small-scale clearing (patches below the 6.25 ha reporting threshold of PRODES) now dominates Amazonian loss, motivating the use of the 30 m Global Forest Change product rather than coarse official clearing inventories in this study.

2.3 Uncontacted peoples: ethics, records, and protection

Research involving uncontacted peoples is bounded by a strict non-contact norm codified in Brazilian policy since 1987 [1] and in the international framework of ILO Convention 169 [22] and UNDRIP [23]. Systematic knowledge therefore comes from indirect evidence: FUNAI's registry of records (114 entries in 2024, across three evidentiary categories) [3], civil-society observatories such as OPI, whose platform mirrors the official categorisation (115 records: 29 confirmed, 26 under study, 60 information-only as of October 2026) [15], and regional compilations [24]. FUNAI's 2024 internal audit documents the administrative fragility of the protection system, including twelve records lacking any territorial protection instrument [3]. Quantitative, territory-level measurement of forest loss around these lands has been absent; this study supplies it without relaxing any ethical constraint.

3 Data

3.1 Territory boundaries

We use the official FUNAI registry of Indigenous Lands (*Terras Indígenas do Brasil — Delimitações Oficiais FUNAI*), exported on 2026-10-29 as 657 polygons with attributes for name, legal stage, modality, and official area. After name-based dissolving (11 duplicate polygons), 646 distinct territories remain. Territory polygons are public policy artefacts; we treat their boundaries as given and discuss positional uncertainty in Section 7.

3.2 Isolated-peoples presence

Presence of uncontacted peoples is established from the OPI observatory (*Observatório para a Proteção dos Povos Isolados e de Recente Contato*), whose platform and GeoServer mirror FUNAI's record system: 115 active records (29 *confirmed*, 26 *under study*, 60 *information*), closely matching the 114 records documented by FUNAI's own audit [3, 15]. OPI further publishes two mapped polygon sets per record: an *occupation area* (*área de ocupação*) and a surrounding *entorno area*. We define a territory as a *TIPI* (Indigenous Land with isolated-peoples presence) if it intersects at least one occupation area by more than 100 ha. This spatial rule yields 81 TIPI; 32 of them are additionally matched by explicit record-to-territory name linkage. The spatial rule is deliberately inclusive: it captures territories that host isolated presence documented by occupation areas even when the record-to-name link is indirect. We report a sensitivity analysis on the stricter name-linked subset (Section 5). The record-coordinate layer is used only for this validation and is never published (Section 8).

3.3 Forest cover change

Forest loss is measured with the Hansen/UMD Global Forest Change dataset, version GFC-2023-v1.11 [16, 17]: 30 m rasters of year-2000 tree cover (*treecover2000*) and annual loss year (*lossyear*, 2001–2023). We define forest as pixels with tree cover above the standard 30% canopy threshold and forest loss as a loss-year pixel under that mask. The study window is therefore 2000–2023. GFC detects gross clearing and industrial loss at 30 m; it does not capture sub-canopy degradation — a limitation for logging fronts but appropriate here, since small clearings dominate recent Amazonian loss [21].

3.4 Accessibility covariates

As accessibility proxies we use OPI's mapped federal highway layer (123 features), railways (676 features), and city/town points (1,054 features) within the Amazon biome, plus the RAISG/OTCA 2024 Amazon biome boundary (698.3 million hectares, validated against published biome areas) [24]. Distances are computed from territory boundaries to the nearest feature in the equal-area reference frame EPSG:6933.

3.5 Official protection status

FUNAI's internal audit report RA-028/2 (2024) supplies the official counts used for context: 114 records, three evidentiary categories, twelve records with no territorial protection, and a staffing deficit of 74 of 156 authorised field posts [3].

Table 1: Data sources. All layers are public; access dates 2026-10-29.

Layer	Source	Version/date	Role
Indigenous Land boundaries	FUNAI (official delimitations)	2026-10-29 export	study units
Isolated-peoples occupation areas and records	OPI GeoServer (WFS)	2026-10-29 export	TIPI definition
Forest cover change 2000–2023	Hansen/UMD GFC v1.11	2023 vintage	response variable
Highways, railways, cities	OPI layers	2026-10-29 export	accessibility
Amazon biome boundary	RAISG/OTCA	2024	study-area mask
Protection audit	FUNAI Audin RA-028/2	2024	policy context

4 Methods

4.1 Study design

The unit of analysis is the Indigenous Land. We employ two complementary comparison structures, following the matched-control and buffer logic standardised in protected-area effectiveness research [18, 19]:

1. **Matched group contrast (RQ1a).** Each TIPI ($n = 81$) is matched 1:2, without replacement, to non-isolated Indigenous Lands from the same state (UF) with the nearest log-area (160 controls; 155 fall within the analysis extent). This controls for state-level governance and for scale effects while comparing otherwise similar tenure categories.
2. **Buffer rings (RQ1b).** For every TIPI we construct two concentric rings around its boundary: 0–10 km and 10–50 km. Rings capture the surrounding pressure regime at two distances and enable a paired within-territory comparison that removes region-level confounding. Rings may include other protected lands, which makes the inside-vs-ring contrast conservative.

Buffers and rings are computed in the equal-area frame EPSG:6933. For rasterisation only, geometries are simplified with a 0.0005° (≈55 m, about two GFC pixels) tolerance, which is negligible relative to ring widths of 40 km; zone areas are always computed from the unsimplified buffers.

4.2 Zonal forest-loss computation

For each territory and zone we count (i) year-2000 tree-cover pixels (canopy >30%) and (ii) loss pixels by year (lossyear 1–23 under the tree mask). The primary response is the *loss share*: the fraction of a zone's year-2000 tree cover lost by 2023. Pixel counts are exact within the native Web-Mercator-free GFC grid (EPSG:4326, 0.00025°); hectares are approximated with a per-window mean-latitude pixel-area factor (maximum ∼1% bias across the latitudinal span of a window) and are used for interpretable magnitudes, while the primary metric is the pixel ratio.

A practical constraint shaped the implementation: GFC GeoTIFF tiles are row-striped with full-width 40,000-pixel rows, so windowed reads fetch whole rows. We therefore stream each required tile in row blocks (1,500 rows × 40,000 columns) and process all candidate territories intersecting each block, fetching only row ranges actually spanned by study territories. The engine is crash-safe (per-tile progress bookkeeping) and covered by unit tests, including a regression test for a block-boundary georeferencing bug that we detected and fixed during development. All 81 TIPI and 155 controls were processed across 20 GFC tiles with no errors; every territory returned valid tree-cover counts.

4.3 Statistics

Primary hypotheses are tested on paired territory-level values:

- **H1.** Median loss share inside TIPI is lower than in each ring (two-sided Wilcoxon signed-rank; paired t reported for completeness).
- **H2.** Ring-0–10 km loss share is negatively associated with distance to the nearest highway and city (Spearman correlation).
- **H3.** Pressure is concentrated: measured by the Gini coefficient and the top-quintile share of absolute 0–10 km ring loss (hectares).

The TIPI-versus-controls contrast uses a two-sided Mann–Whitney U test with the rank-biserial effect size. We report Cohen's d_z and rank-biserial r for paired tests. Family-wise error across the five primary/secondary tests is controlled with Holm–Bonferroni adjustment. No Gaussian assumptions are made for inference; means are reported alongside medians for context only.

4.4 Reproducibility

The entire analysis — data acquisition, territory construction, zonal computation, statistics, and figures — is scripted (Python 3.12; `geopandas`, `rasterio`, `numpy`, `pandas`, `scipy`), seeded (20260101) where randomness exists (matching), and re-runnable end-to-end with one command from the open repository. Machine-readable outputs (CSV/JSON) are emitted at every stage, and every number in this paper is traceable to a result file.

5 Results

5.1 Descriptive overview

The 81 TIPI span 89.3 million hectares, of which 86.5 million hectares (97%) carried tree cover above 30% in 2000; the median territory held 403,600 ha of tree cover (Table 2). Forest loss

inside TIPI over 2000–2023 totalled 1.53 million hectares, against 2.88 million hectares in the immediately surrounding 0–10 km rings and 16.3 million hectares in the 10–50 km rings.

Table 2: Forest-cover loss 2000–2023 by zone and group. Loss share is the fraction of year-2000 tree cover (canopy >30%) lost; IQR in brackets. "Controls" are matched non-isolated Indigenous Lands (inside zone only shown; $n = 155$).

Group	Zone	n	Median loss share	Mean	Total loss (ha)
TIPI	Inside	81	1.01% [0.29–3.45]	3.13%	1,533,682
TIPI	0–10 km ring	81	3.66% [0.83–14.5]	9.98%	2,880,999
TIPI	10–50 km ring	81	5.30% [1.22–17.9]	10.98%	16,344,610
Controls	Inside	155	2.51% [0.92–7.29]	6.75%	1,006,569

5.2 RQ1: inside versus rings and matched controls (H1)

Forest loss inside TIPI is far lower than immediately outside (Figure 1). The median territory lost 1.01% of its year-2000 tree cover internally, versus 3.66% in its 0–10 km ring and 5.30% in the 10–50 km ring. The paired differences are strongly significant and large: Wilcoxon signed-rank $W = 274$, $p = 6.7 \times 10^{-11}$ (inside vs. 0–10 km; Holm-adjusted $p = 2.7 \times 10^{-10}$; Cohen's $d_z = -0.72$; rank-biserial $r = -0.83$) and $W = 149$, $p = 1.1 \times 10^{-12}$ (inside vs. 10–50 km; adjusted $p = 5.5 \times 10^{-12}$; $d_z = -0.83$; $r = -0.91$). On average, the immediate ring lost 3.2 times more of its tree cover than the territory it surrounds; the wider ring lost 3.5 times more.

Relative to matched non-isolated Indigenous Lands, TIPI also fared better: median inside loss share 1.01% versus 2.51% for controls (Mann–Whitney $U = 4252$, $p = 4.8 \times 10^{-5}$; Holm-adjusted $p = 9.5 \times 10^{-5}$; rank-biserial $r = -0.32$). The controls' own 0–10 km rings lost a median 11.0%, indicating that the region around a randomly matched Indigenous Land is under roughly three times the pressure of the region around a TIPI — consistent with the remoteness of TIPI documented in Section 5.4.

A sensitivity analysis restricted to the 32 territories with explicit record-to-territory linkage (the strictest definition of hosting uncontacted peoples) strengthens every contrast: median inside loss 1.33% versus 9.77% (0–10 km) and 13.6% (10–50 km) in rings (Wilcoxon $W = 47$, $p = 9.9 \times 10^{-6}$ and $W = 9$, $p = 1.5 \times 10^{-8}$).

5.3 Annual trajectories

Inside loss was negligible and stable through 2015 (below 0.08% of tree cover per year), then rose sharply in 2016 (0.41%) — a year combining a severe drought-fire season with national environmental-governance deterioration — and remained elevated through 2023 (0.06–0.18% per year). Aggregate inside loss more than tripled between the decades: 3.75 million loss pixels in 2001–2010, 10.5 million in 2011–2018, and 5.9 million in the five years 2019–2023. Ring trajectories mirror the same 2016–2017 peak (up to 0.73% per year in the 10–50 km ring), indicating that the pressure regime around these territories intensified across the board rather than only at their edges (Figure 3).

5.4 RQ2: accessibility (H2)

Ring-level pressure around TIPI is *not* significantly associated with distance to the nearest mapped federal highway (Spearman $\rho = -0.16$, $p = 0.16$; Holm-adjusted $p = 0.16$), but it is strongly associated with distance to the nearest city ($\rho = -0.46$, $p = 1.8 \times 10^{-5}$; adjusted $p = 5.4 \times 10^{-5}$; Figure 4). The null highway result is informative rather than merely negative: several of the highest-pressure territories (e.g., Uru-Eu-Wau-Wau in Rondônia) sit near trunk roads, but many heavily pressured territories are instead exposed to settlement mosaics, informal roads,

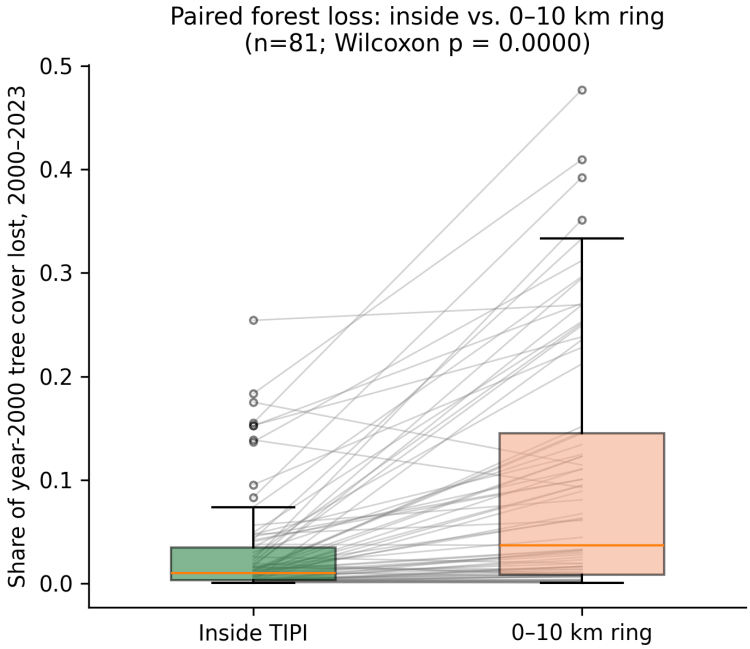


Figure 1: Paired forest-loss shares (fraction of year-2000 tree cover lost 2000–2023) inside TIPI versus their 0–10 km rings. Grey lines connect paired values for each of the $n = 81$ territories.

and riverine access not captured by the federal-highway layer, while some highway-adjacent territories retained low loss — indicating that legal protection status and enforcement, not road proximity alone, condition outcomes. City distance captures the diffuse settlement frontier that the highway layer misses.

5.5 RQ3: concentration and hotspots (H3)

Pressure is highly concentrated. The Gini coefficient of absolute 0–10 km ring loss across the 81 TIPI is 0.63; the top quintile of territories holds 65.7% of aggregate ring loss, and the ten highest-pressure territories alone account for 48.5% (Table 3). The hotspot list is strikingly consistent with documented invasion crises: Uru-Eu-Wau-Wau (Rondônia) leads with 222,000 ha of ring loss and 35% of its own tree cover lost — a territory whose invasions and isolated-groups murders were nationally reported — followed by the Xingu, Kayapó, Baú and Menkragnoti lands (soy/land-grabbing frontiers in Mato Grosso and Pará) and Arariboia (Maranhão), where ring loss reached 48% of tree cover. Conversely, flagship protected territories such as Massaco (1.25% inside loss; the first Indigenous Land demarcated exclusively for uncontacted peoples) and Rio Omerê (0.59% inside loss, surrounded by a 26.6% ring) show that strong protection can hold even under high surrounding pressure.

6 Discussion

6.1 Interpretation

Three findings stand out. First, the lands of uncontacted peoples are, quantitatively, the best-buffered Indigenous Lands in the Brazilian Amazon: 1.0% median internal loss over 24 years, five times below the surrounding rings (median basis), and 2.5 times below matched non-isolated controls. Given that these territories host the only human populations for whom forest loss translates directly into existential risk, this asymmetry is reassuring — but it is a statement about the present, not the future, because the second finding shows the buffer is thinning.

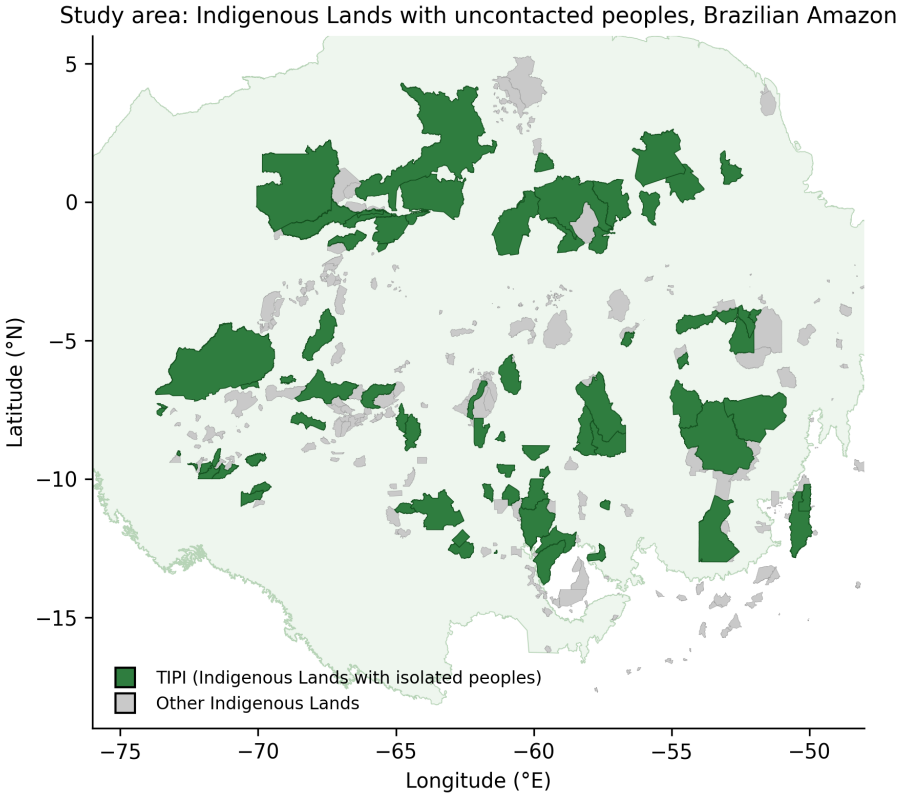


Figure 2: Study area. Green polygons are the 81 Indigenous Lands with isolated-peoples presence (TIPI); grey polygons are other official FUNAI Indigenous Lands within the analysis extent. Map coordinates are approximate; no settlement or record locations are shown.

Second, the trajectory: internal loss tripled after 2011 and jumped in 2016, tracking the national post-2012 deforestation surge and the 2015–2016 drought-fire season. Ring pressure (up to 0.73%/yr in 2017) describes a closing frontier. The FUNAI audit's administrative findings — twelve records with no territorial protection, a 47% staffing deficit, outdated registries — read very differently against this quantitative backdrop: the apparatus protecting these lands was weakening exactly while pressure accelerated [3].

Third, pressure concentrates in a handful of territories that are already known crisis sites. This concentration is actionable: a protection policy that prioritised the top-quintile territories (65.7% of aggregate surrounding loss) would address most of the threat surface with a fraction of the effort — and the audit's own diagnosis (under-staffed Field Protection Coordinations) provides the mechanism through which such prioritisation should flow [3].

The accessibility result adds nuance to the roads literature [10, 11]: trunk highways alone did not predict ring pressure here, whereas city proximity did. The likely mechanism is that the federal-road layer misses the informal road networks and settlement mosaics that radiate from towns — the same infrastructure that official clearing inventories under-detect [21].

6.2 Comparison with prior work

Our inside-vs-buffer asymmetry (3–3.5×) is larger than the protection factors typically reported for Indigenous lands in general, which matched-control designs place in the 2–3× range [5, 18, 19]. Two mechanisms plausibly combine: genuine protection effects of isolation (no market access, no demand for clearing inside), and residual remoteness confounding that matching on area and state does not fully remove. We therefore deliberately avoid a causal claim (Section 7). The hotspot consistency with BR-319 and Rondônia frontiers [12] and with the basin-scale finding

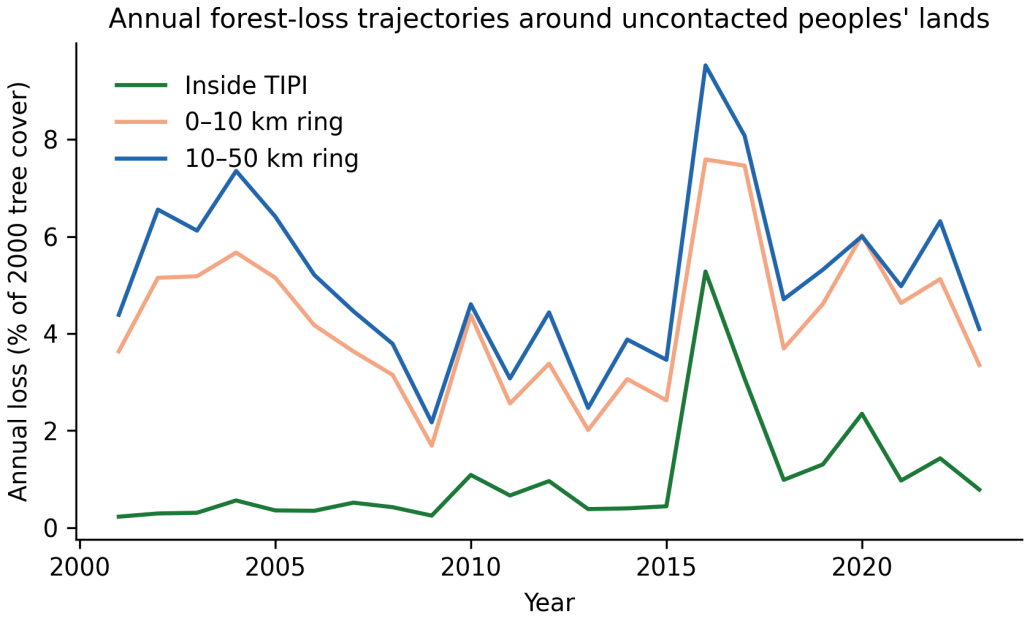


Figure 3: Annual forest loss as a percentage of year-2000 tree cover, aggregated across the 81 TIPI for each zone.

Table 3: Ten highest-pressure TIPI by absolute forest loss in the surrounding 0–10 km ring, 2000–2023.

Territory	State	Area (M ha)	Ring loss (ha)	Inside loss (%)
Uru-Eu-Wau-Wau	RO	1.87	222,349	35.1
Parque do Xingu	MT, PA	2.64	201,849	27.1
Kayapó	PA	3.28	184,581	25.3
Baú	PA	1.54	153,702	23.5
Menkragnoti	PA, MT	4.93	123,626	11.1
Arariboia	MA	0.41	112,000	47.7
Cachoeira Seca	PA	0.73	108,125	22.8
Nambikwara	MT	1.00	105,702	21.2
Parque do Aripuanã	MT, RO	1.60	101,597	15.2
Yanomami	RR, AM	9.49	83,290	3.0

that Indigenous and protected lands are far less impacted than unprotected ones [14] supports external validity.

7 Limitations and Threats to Validity

1. **Remoteness confounding.** TIPI are located in less accessible regions by construction of the protection system itself; matched controls and rings mitigate but do not eliminate this. Our claims are comparative descriptions, not causal estimates.
2. **Registry uncertainty.** The official territory registry and the isolated-peoples record system contain documented inconsistencies [3]; our TIPI definition (occupation-area intersection, 81 lands; name-linked subset, 32) is inclusive and sensitivity-tested, but some territories may be misclassified in either direction.
3. **Boundary and detection error.** Territory polygons carry positional uncertainty; GFC does not detect sub-canopy degradation (selective logging, fire under canopy), which likely undercounts pressure — plausibly more inside territories, where logging is covert, than in

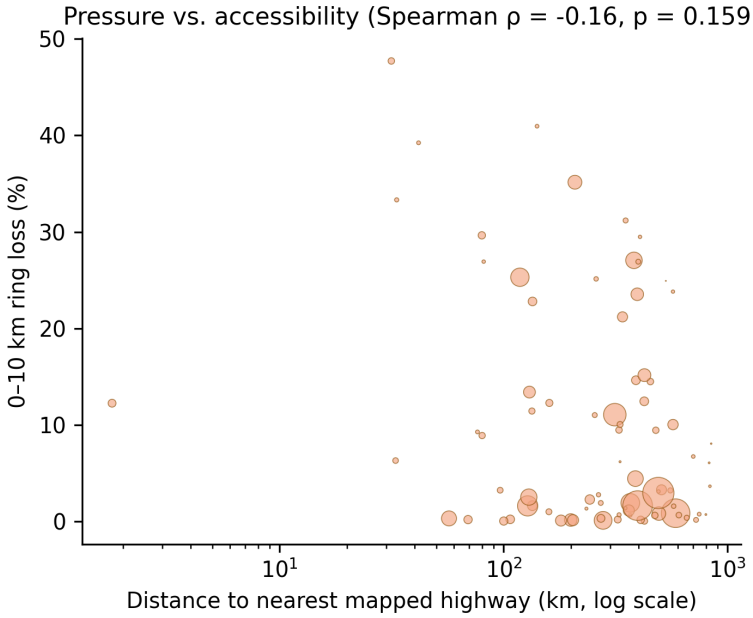


Figure 4: 0–10 km ring loss share versus distance to the nearest mapped highway (log scale); point area proportional to territory size.

clear-cut rings. The 2016 inside-loss spike, coinciding with drought-fire years, may partly reflect fire detected through canopy.

4. **Buffer overlap.** Rings may include other Indigenous Lands or protected areas, which biases the inside-vs-ring contrast towards zero (conservative).
5. **Spatial pseudo-replication.** Analyses are aggregated at territory level ($n = 81$); pixels are never treated as independent. Ring autocorrelation across neighbouring territories is not modelled.
6. **Metric approximation.** Hectare conversions use a per-window mean-latitude pixel-area factor ($\leq$1% bias); primary inference uses exact pixel shares.
7. **No ground validation.** By design (Section 8), no field verification of loss events inside territories was possible.

8 Ethical Considerations

This study was designed around a binding constraint: uncontacted peoples cannot consent to be studied, and any approach — physical or informational — that increases contact risk is unacceptable. Brazil's no-contact policy (in force since 1987), ILO Convention 169, and UNDRIP frame isolation as a choice that outsiders must respect [1, 22, 23]. Accordingly:

1. **No contact.** The study uses exclusively remote, public, aggregate data. No field work, overflights, interviews, or interactions of any kind are proposed or implied.
2. **No location disclosure.** Coordinates of isolated-peoples records and occupation areas are used only internally to link records to official territory boundaries. No figure, table, or repository artefact publishes settlement or record locations; all outputs are aggregated to Indigenous-Land boundaries (public policy artefacts) or to regional summaries.
3. **Respectful framing.** We describe peoples, not "tribes", and make no claims about lifeways; the object of measurement is land and threat, not people.
4. **Data protection.** The raw record layer is excluded from the repository by default.

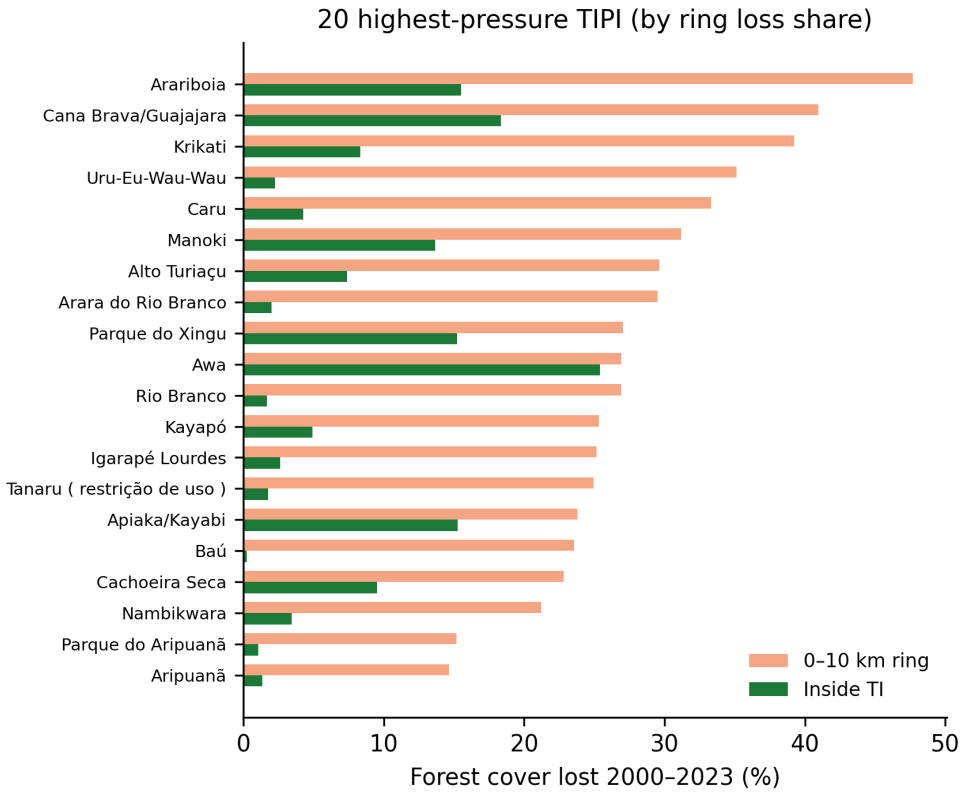


Figure 5: Twenty highest-pressure TIPI ranked by 0–10 km ring loss share (sand), with inside loss share (green) overlaid.

9 Conclusions and Future Research

Using only open data, we provide the first territory-level quantitative assessment of forest-loss pressure on Brazil's Indigenous Lands hosting uncontacted peoples. Between 2000 and 2023 these 81 lands lost 1.0% of their year-2000 tree cover internally — five times less than their surroundings and 2.5 times less than matched non-isolated Indigenous Lands — confirming their exceptional buffering role. But the buffer is thinning: internal loss tripled after 2011, pressure concentrates sharply (top quintile: 65.7% of aggregate ring loss) in territories that are already documented crisis sites, and the state's protection apparatus was audited in 2024 as under-staffed and leaving twelve records wholly unprotected.

Future work should pursue four directions: (i) causal designs exploiting restriction-of-use ordinances and demarcation timing (staggered difference-in-differences over the territory panel) to separate protection from remoteness; (ii) extension to Peru, Colombia, and Ecuador via RAISG's isolated-peoples layers, testing external validity beyond Brazil; (iii) integration of fire and degradation products (active-fire archives, canopy-height change) to capture sub-canopy pressure invisible to gross-clearing products; and (iv) a systematic, ethics-preserving assessment of the *protection-coverage gap* — records without any territorial instrument — by joining the audited record registry to protection-layer geometry.

Declarations

Funding. None. **Conflicts of interest.** None declared. **Data and code availability.** All data are public (Table 1); all code, configuration, result files, and the audit trail are available in the open repository accompanying this preprint. **AI-assistance disclosure.** AI tools were used for code development and manuscript drafting under the author's direction; the author

verified all results, retains full responsibility for the work, and discloses this assistance in line with venue policy. **Ethics.** No human subjects, no animals, no field work; see Section 8.

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